Capital Franchise Supply Chain Management System
Software Requirement Specification

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Approval Page

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1. Introduction

1.1 Company Overview

Capital Corp. is a leading franchising organization that specializes in providing products, infrastructure, and equipment to support its franchise network globally. With more than 100 franchises located in strategic areas, Capital Corp.'s mission is to deliver efficient supply chain solutions that enhance the productivity and competitiveness of each franchise.

The **Capital Franchise Management System (CFMS)** is designed to integrate core operational processes for franchises, including inventory management, warehouse tracking, order handling, and access permission management. Additionally, the system supports advanced features like AI booking for POS offline to improve user ordering experiences.

1.2 Solution Concept

The CFMS offers a centralized digital platform to support franchise operations. Its key goals include:

- **Empower AI Booking for POS:** Enables customers to place orders via voice even in offline scenarios.
- **Supporting Franchise Setup:** Enabling core subscription templates to be defined and cloned for rapid deployment.
- **Warehouse Management Integration:** Overseeing the purchasing, exporting, and inventory tracking in real-time, providing franchise-specific insights.

2. Business Requirements

2.1 Actors

The system includes Administrator, Staff, and Customer roles, each with specific access rights to ensure secure and efficient functionality.

Actor	Description
Administrator	A user with full control over the entire system. Administrators can manage user accounts and permissions through the IAM Service, oversee payment processing and payment methods via the Payment Service, and configure promotional campaigns using the Promotion Service. They also have access to manage inventory through the Inventory Management Service, oversee product management via the Product Service, and monitor overall system performance through the Reporting Service.
Manager	A user responsible for overseeing specific operational aspects of the business. The Manager interacts with the Shift Management Service to create and manage employee shifts and schedules and Reporting Service to generate and review operational reports concerning sales, inventory, and employee performance. They ensure stock levels through the Inventory Management Service. The Manager makes critical decisions based on insights gathered from these services.
Staff	A coffee shop employee who manages the daily operations of the shop. Staff interact with the Order Management Service to view and update order statuses and the Delivery Management Service to notify customers when orders are ready. Under the guidance of Managers, they ensure smooth operational workflows.
Customer	A company employee who uses the system to browse the menu, place orders, and track their order status. Customers interact primarily with the IAM Service for authentication and the Order Management Service to place and track their orders. They may also engage with the Loyalty Service to earn and redeem loyalty points, as well as take advantage of promotions managed via the Promotion Service. Customers are focused on enhancing their experience through ease of use and access to relevant offers.

2.2 User case diagram of the system

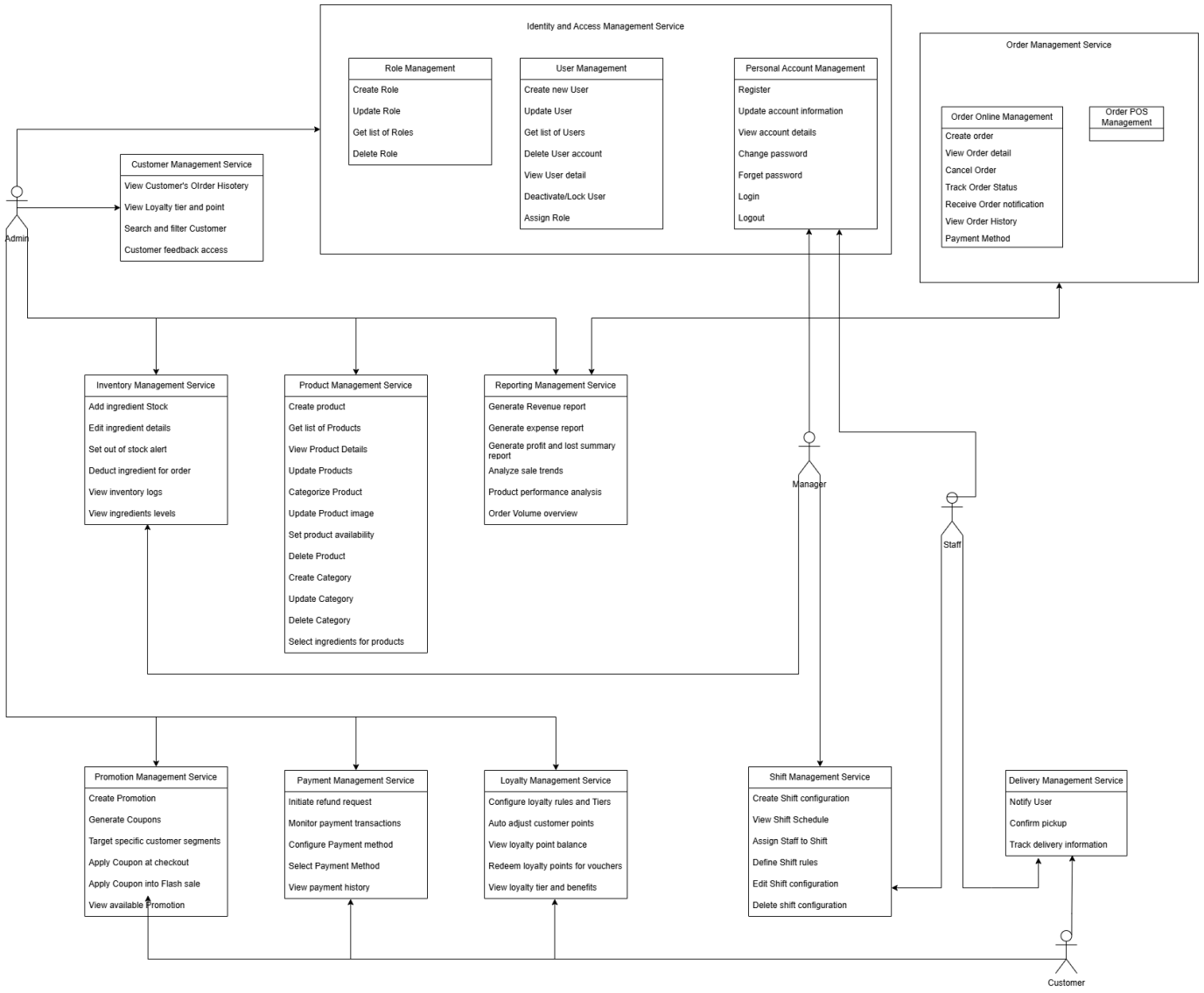


Figure 1: Use Case Diagram

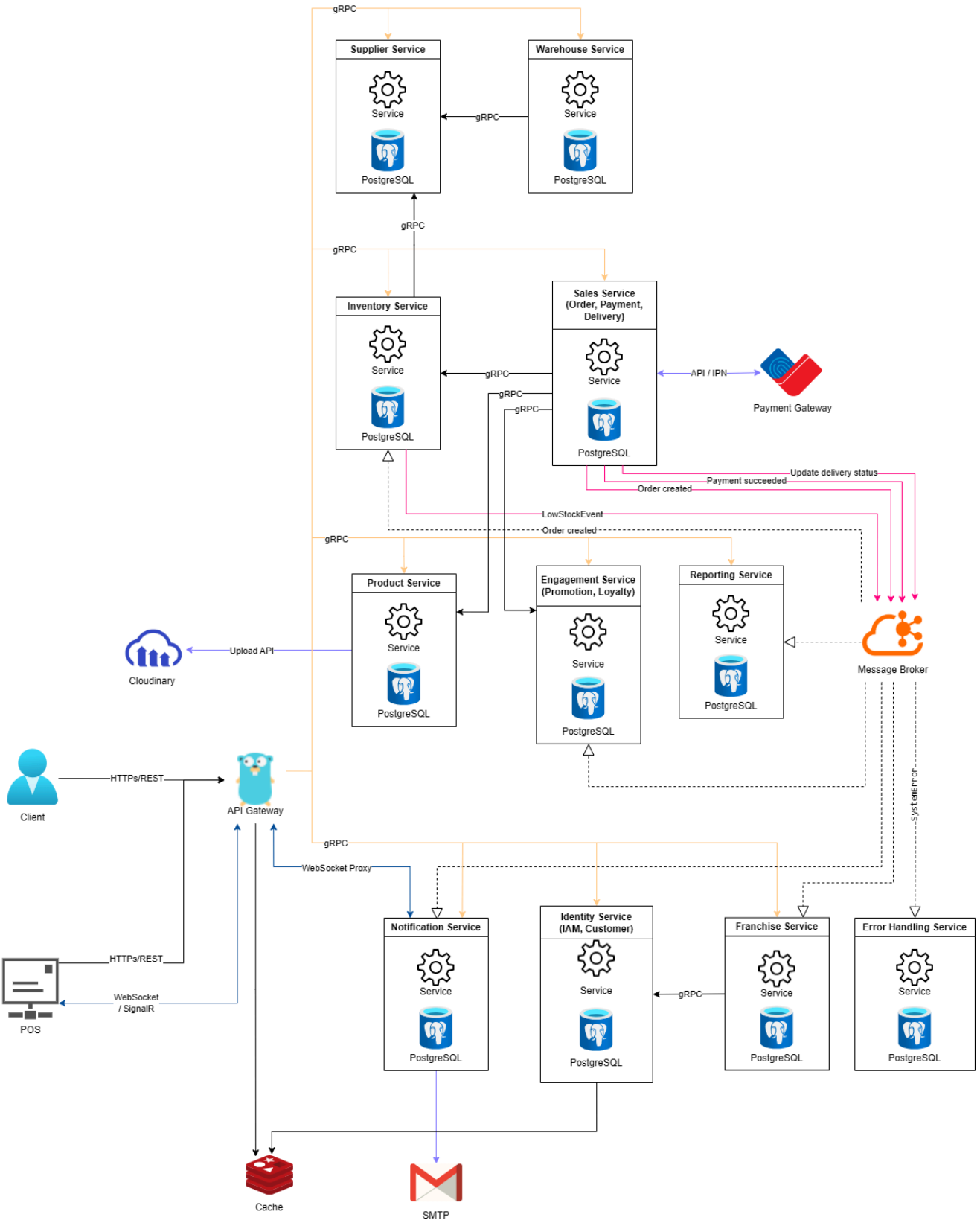


Figure 2: High Level Design

2.3 Configurations List

Configuration Name	Description	Default Value
Lockout Policy	This security feature activates when a user enters an incorrect password multiple times. It temporarily locks the account to prevent unauthorized access and safeguard sensitive information.	5 times
Session Timeout	Determines the duration a user session remains active. After a set period of inactivity, the session automatically expires to maintain security and prevent unauthorized access.	30 minutes
Password Policy	Enforces password complexity requirements, expiration timelines, and reset procedures to strengthen security.	8–12 characters
Auto Deactivation	Once an account surpasses the inactivity threshold, it is automatically disabled to prevent unauthorized access.	60 days
Inventory Threshold	The minimum stock level at which the system should trigger a low inventory alert to notify staff.	10 units
Max Concurrent Orders	Sets the maximum number of active orders the system can process concurrently to manage load and kitchen resources effectively.	50 orders

Pickup Interval	The length of time per available pickup slot that users can select when placing orders.	15 minutes
Flash Sale	Able to hold hundreds of request from user	100+ redeem request

2.4 Privileges List

Category	Privilege	Description	Admin	Manager	Staff	Customer
User & Role Mgmt	View user	Access list of users and profile details.	X	X		
	Create user	Add a new user to the system.	X			
	Modify user	Update existing user information.	X			
	Delete user	Remove a user account.	X			
	View role	View available roles and privileges.	X	X		

	Create role	Create new custom roles.	X			
	Update/Delete role	Edit or remove existing roles.	X			
Product & Inventory	View product	View product catalog and details.	X	X	X	X
	View inventory	View current stock levels and logs.	X	X	X	
	Create product	Add new products (details/images).	X	X		
	Update/Delete product	Modify or remove product entries.	X	X		
	Categorize product	Assign products to categories.	X	X		

	Add ingredient stock	Add new stock to inventory.	X	X	X	
	Edit ingredient detail	Update ingredient info/supplier.	X	X		
	Set out of stock alert	Set alerts for low inventory.	X	X	X	
Orders & Sales	Create order	Place a new order.	X	X	X	X
	View order / detail	Access order history and details.	X	X	X	X
	Update Order Status	Change status as order processes.	X	X	X	
	POS Order Entry	Enter orders via Point of Sale.			X	
	Edit Order	Modify order before preparation.	X	X	X	

	Cancel order	Cancel a previously placed order.	X	X		X
	View Live Orders	Real-time visibility of active orders.	X	X	X	
Reporting & Analytics	Access reporting	General access to system metrics.	X	X		
	Revenue/Expense/ P&L	Generate financial performance reports.	X	X		
	Analyze sales trends	Insights into sales performance.	X	X		
	Analyze customer behavior	View buying patterns/data.	X	X		
Loyalty & Promotions	Configure loyalty	Set up rules, tiers, and benefits.	X	X		

	Adjust customer points	Manually update loyalty points.	X	X		
	Create promotions	Set up new business promotions.	X	X		
	Generate coupons	Create/distribute discount coupons.	X	X		
	Monitor effectiveness	Analyze promotion performance.	X	X		
	Apply coupon	Redeem coupons during checkout.				X
	Create Campaign	Create a campaign that has vouchers to redeem	X			
Logistics & Delivery	View delivery	Monitor delivery details/personnel.	X	X	X	

	Assign Delivery Staff	Assign specific staff to tasks.	X	X		
	Track Delivery Info	Monitor statuses and details.	X	X	X	X
	Confirm Pickup	Confirm order pickup completion.	X	X	X	
Shift Management	Create/Edit Shift	Setup and modify employee shifts.	X	X		
	View Shift Schedule	Access scheduled shifts.	X	X	X	
	Assign Staff	Assign members to specific shifts.	X	X		
	Choose Shift	Choose shift			X	
Payments	Monitor transactions	Access payment details/oversight.	X	X		

	Configure methods	Manage accepted payment methods.	X			
	View payment history	Access past payment transactions.	X		X	

3. Functional Requirements

3.1 Identity and Access Management (IAM)

3.1.1 User Management

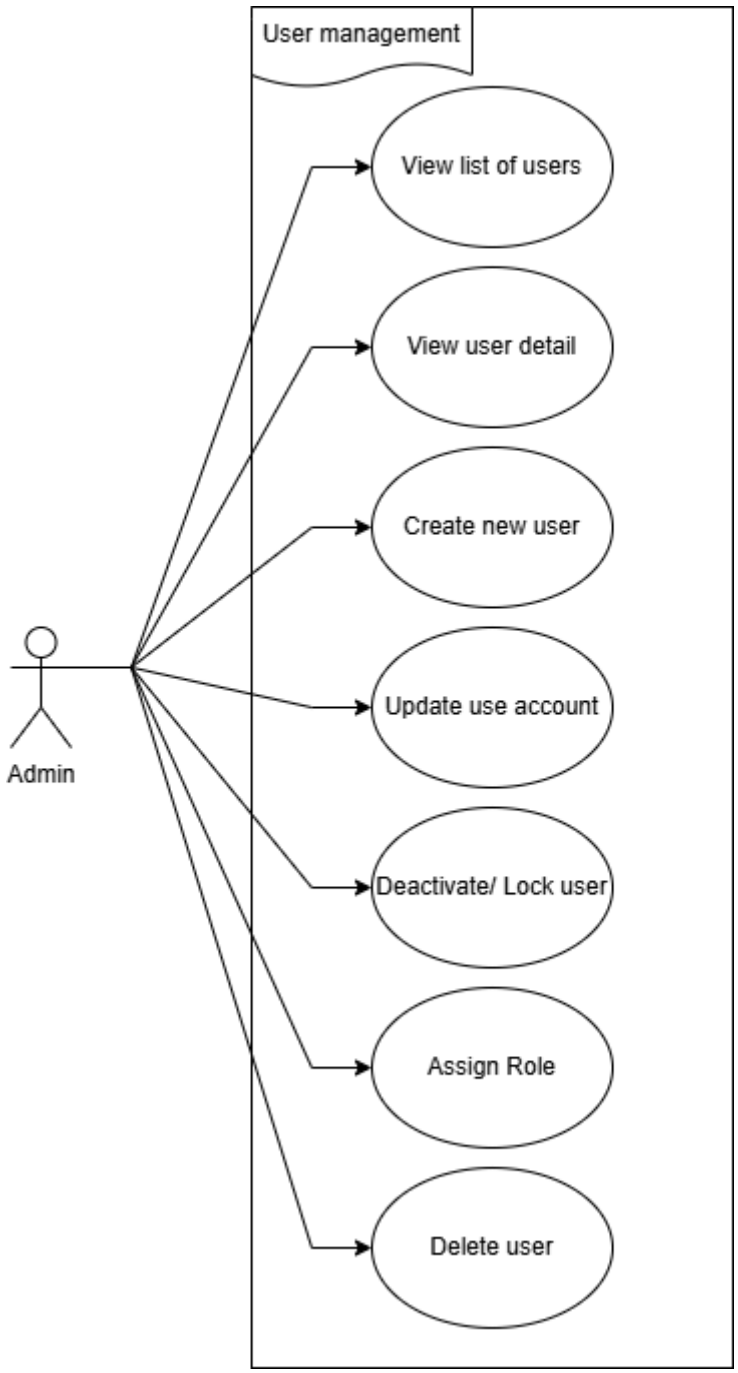


Figure 3: User Management use case diagram

Users Management contains specific 7 features: Create new User, Update User account, Get List of Users, Delete User account, View User details, Deactivate/ Lock User, Assign Role.

3.1.2 Permission Management

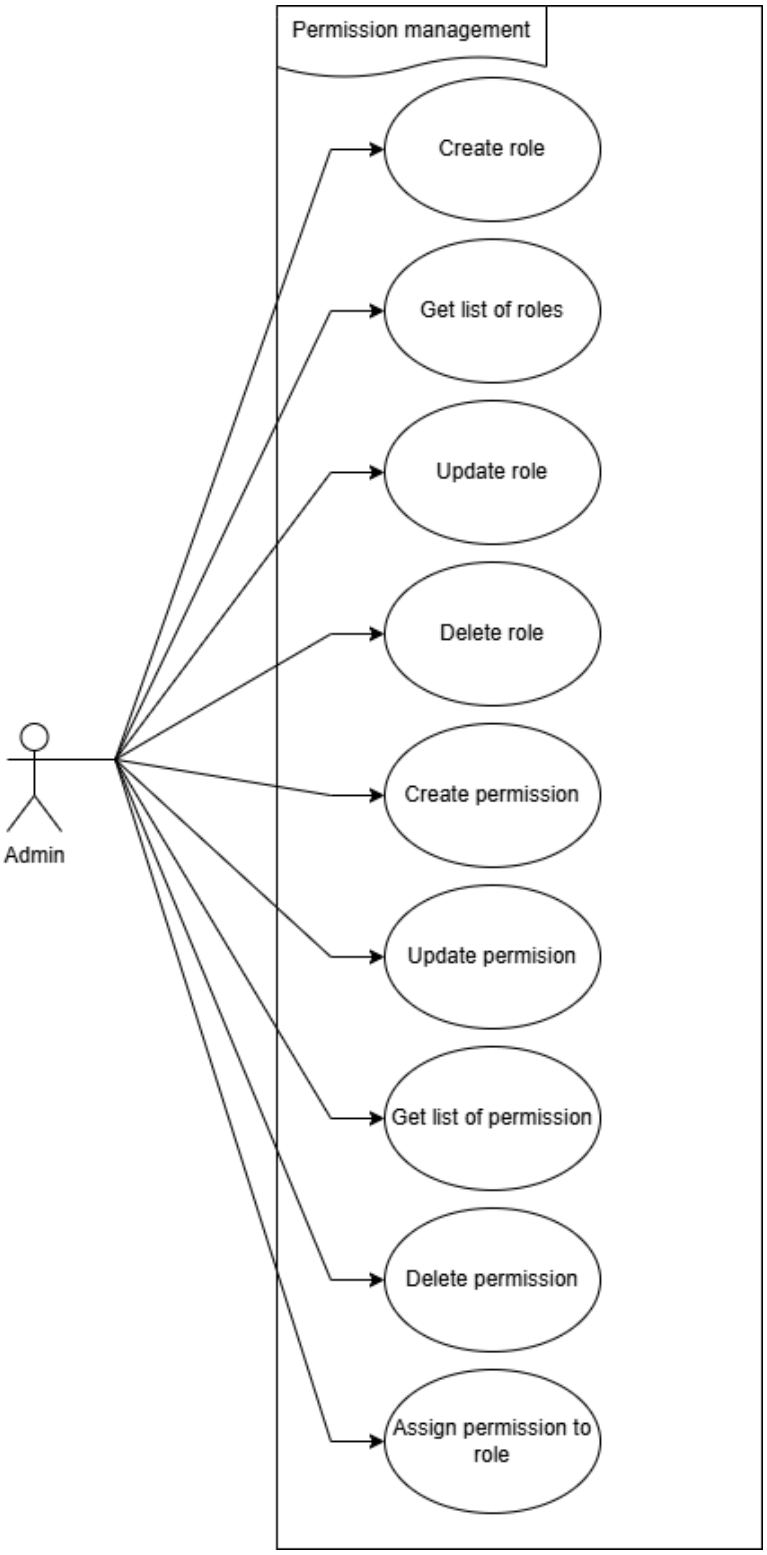


Figure 4: Permission Management use case diagram

Permission Management contains specific 9 features: Create Role, Update Role, Get List of Roles, Delete Role, Create Permission, Update Permission, Get List of Permissions, Delete Permission, Assign Permission to Role

3.1.3 Personal Account

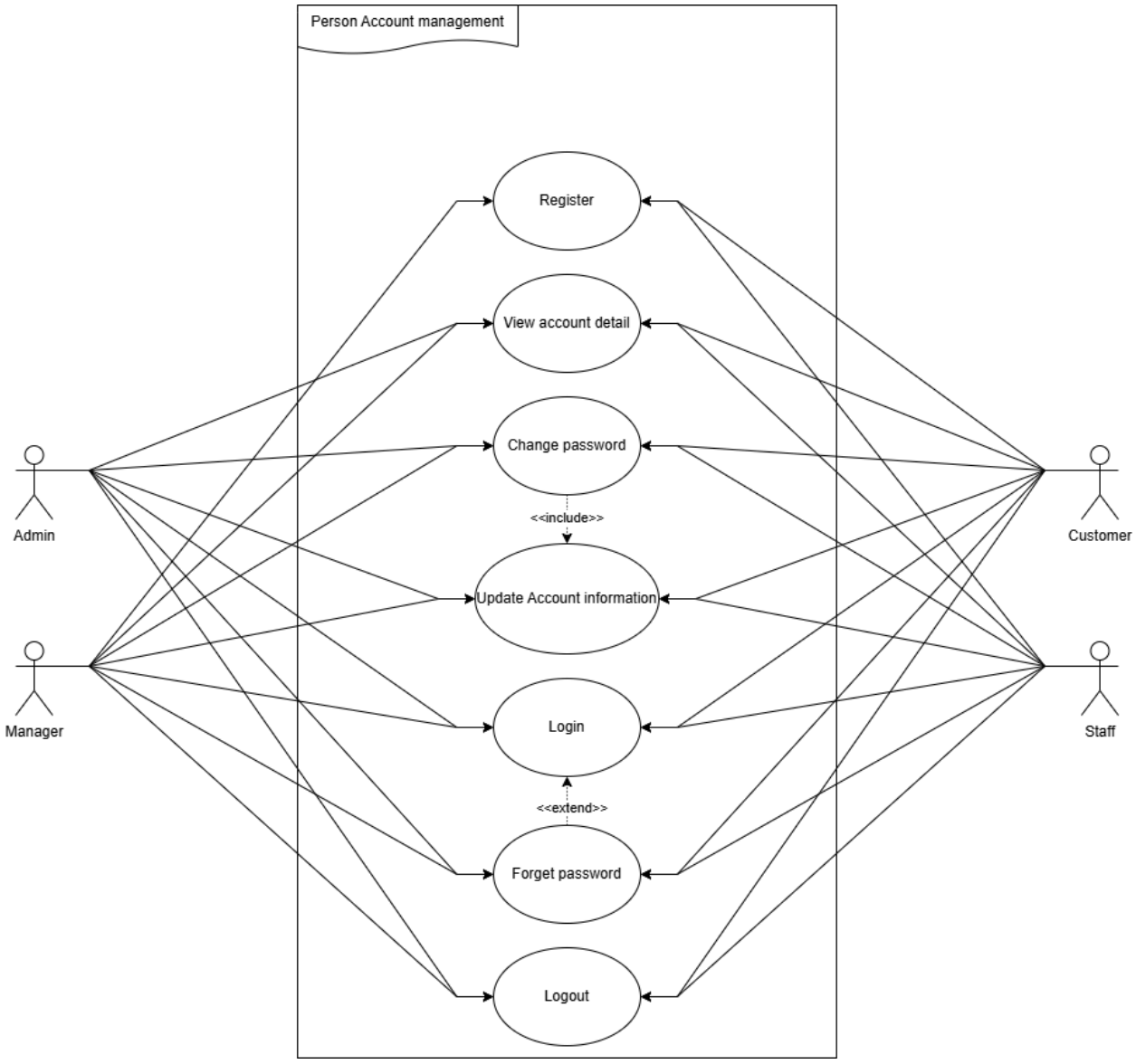


Figure 5: Personal Account Management use case diagram

Personal Account Management contains specific 7 features: Register, Update Account information, View Account details, Change Password, Forget Password, Login, Logout.

3.2 Operations Services

3.2.1 Customer Management

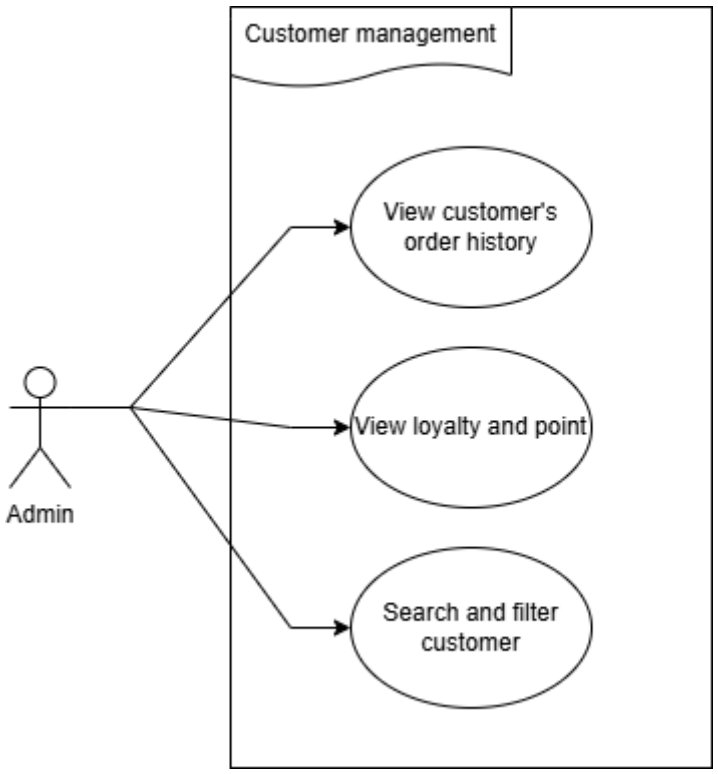


Figure 6: Customer Management use case diagram

Customer Management contains specific 4 features: View Customer's Order History, View Loyalty Tier and Points, Search and Filter Customer, Search and Filter Customer

3.2.2 Reporting Management

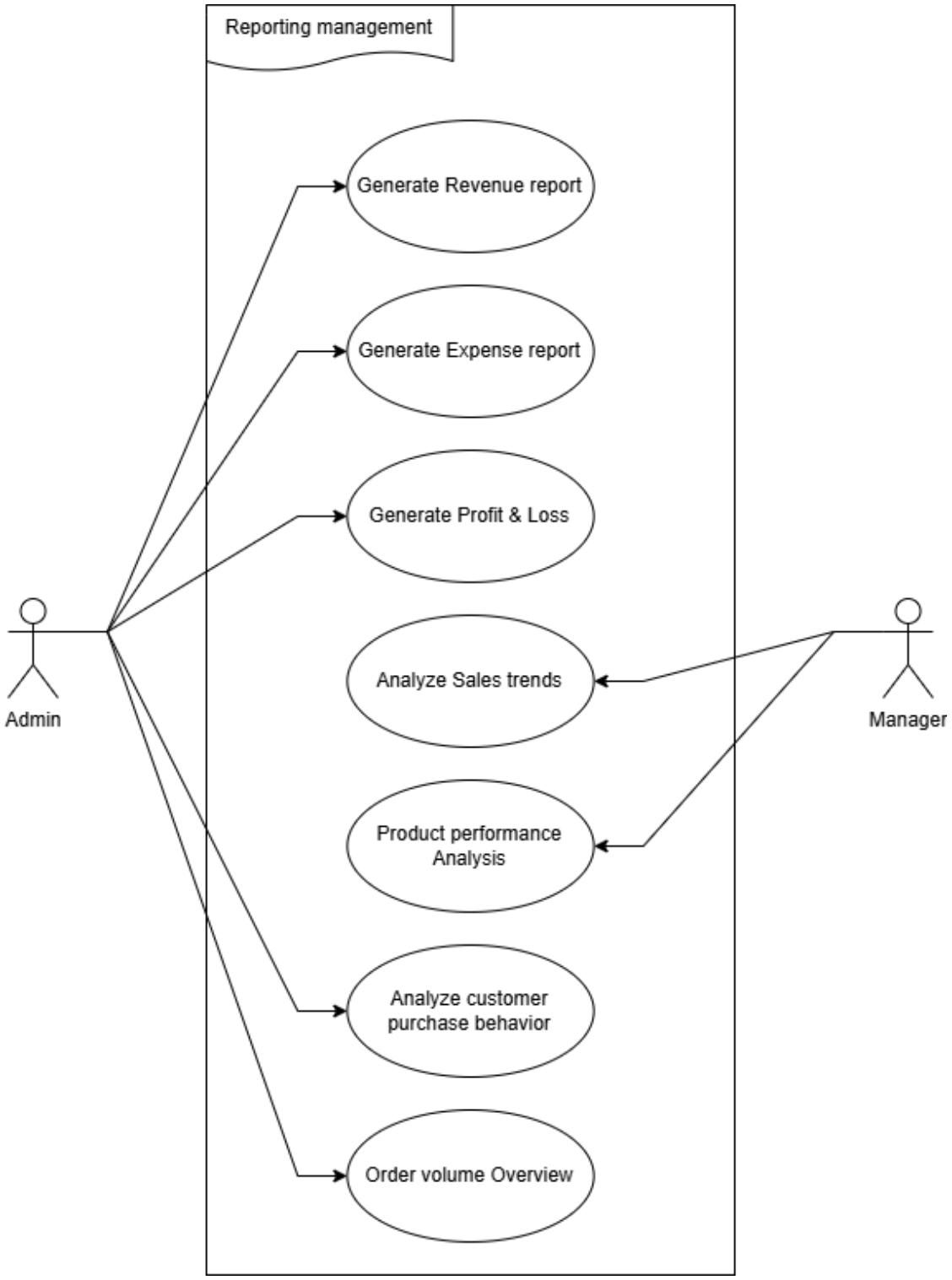


Figure 7: Reporting Management use case diagram

Reporting Management contains specific 7 features: Generate Revenue Report, Generate Expense Report, Generate Profit and Loss Summary, Analyze Sales Trends, Product Performance Analysis, Analyze Customer Purchase Behaviour, Order Volume Overview.

3.2.3 Loyalty Management

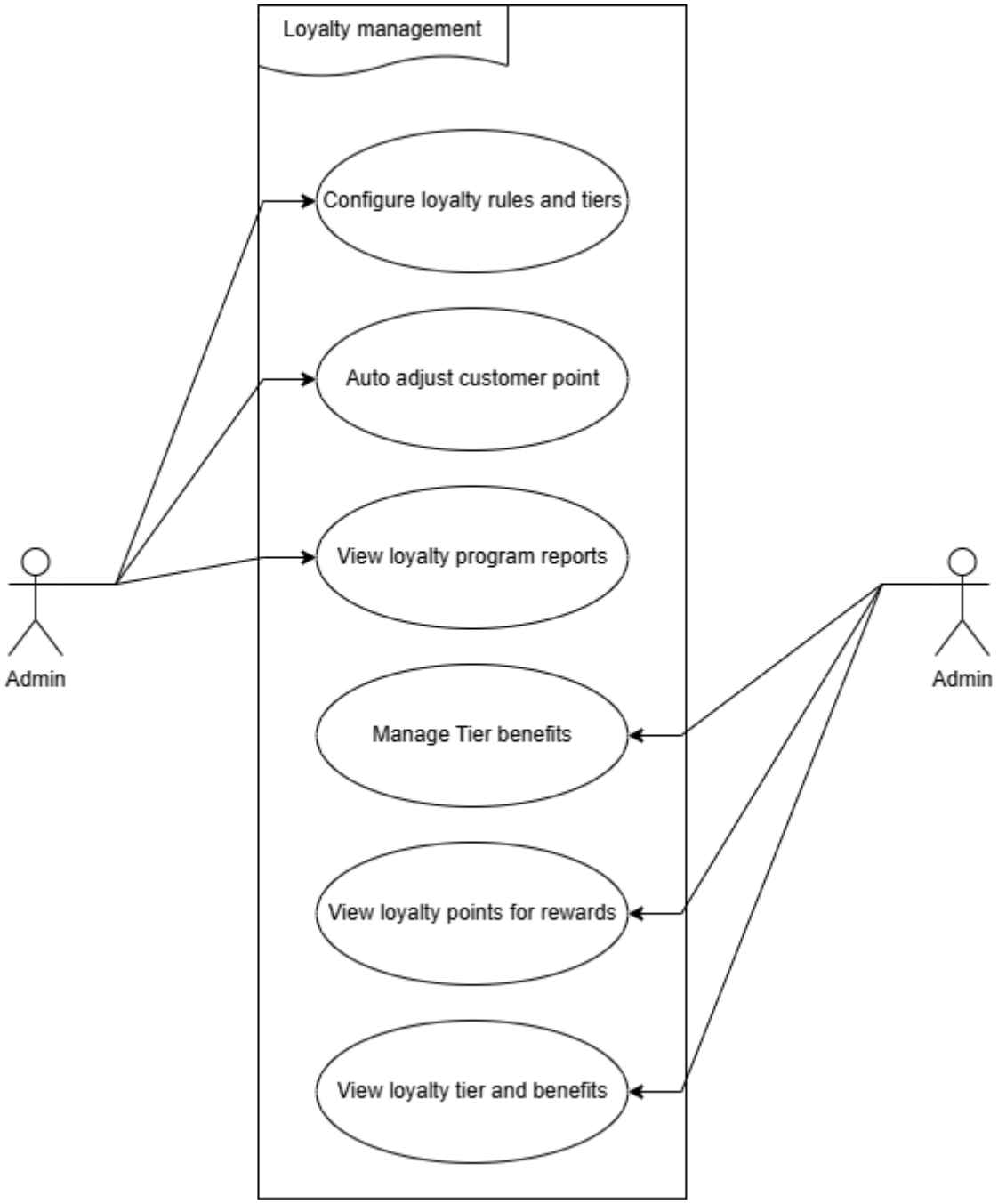


Figure 8: Loyalty Management use case diagram

Loyalty Management contains specific 7 features: Configure Loyalty Rules and Tiers, Manually Adjust Customer Points, View Loyalty Program Reports, Manage Tier Benefits, View Loyalty Points Balance, Redeem Loyalty Points for Rewards, View Loyalty Tier and Benefits

3.2.4 Promotion Management

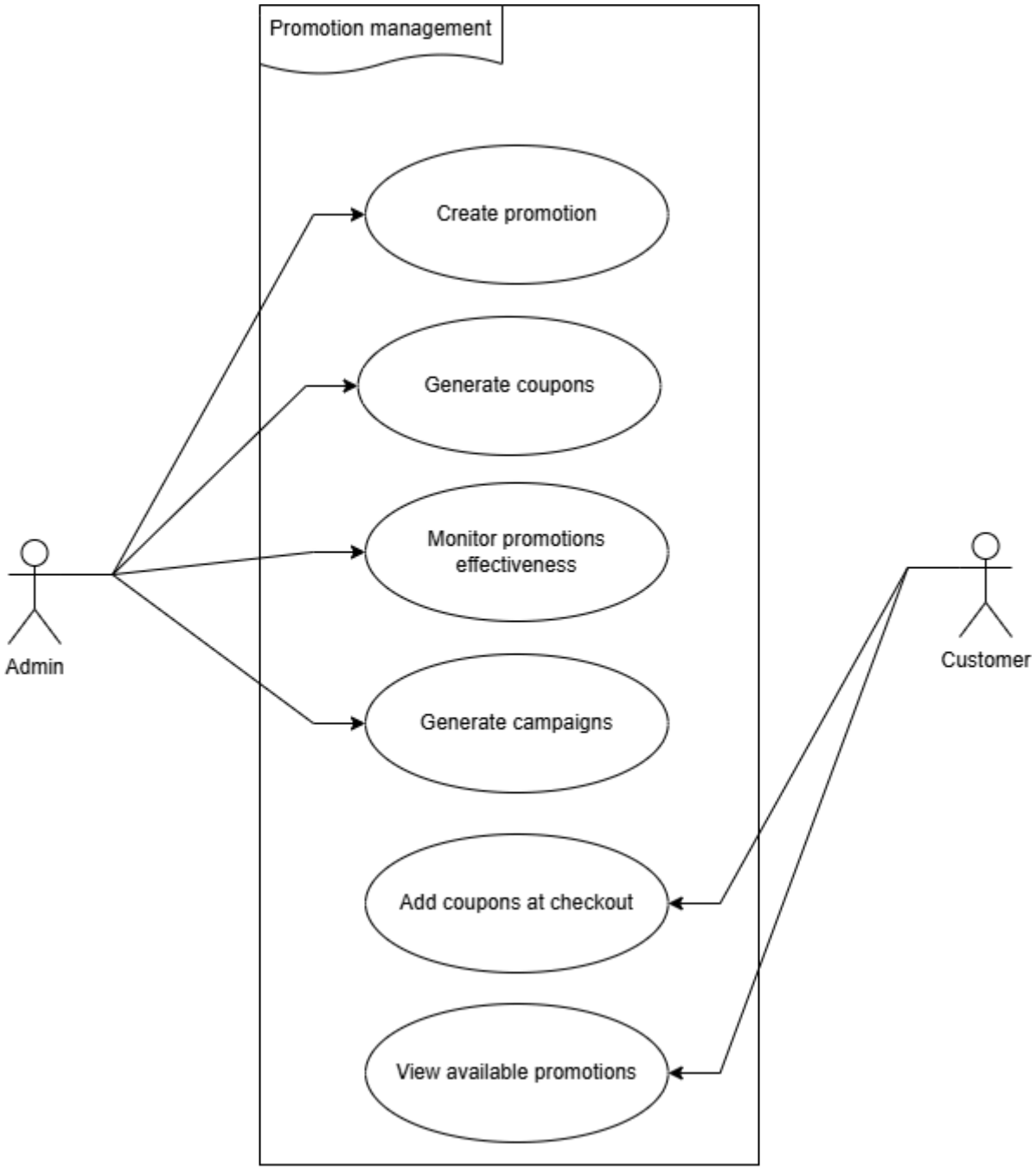


Figure 9: Promotion Management use case diagram

Promotion Management contains specific 7 features: Create Promotions, Generate Coupons, Generate campaigns, Monitor Promotion Effectiveness, Apply Coupon at Checkout, View Available Promotions

3.2.5 Payment Management

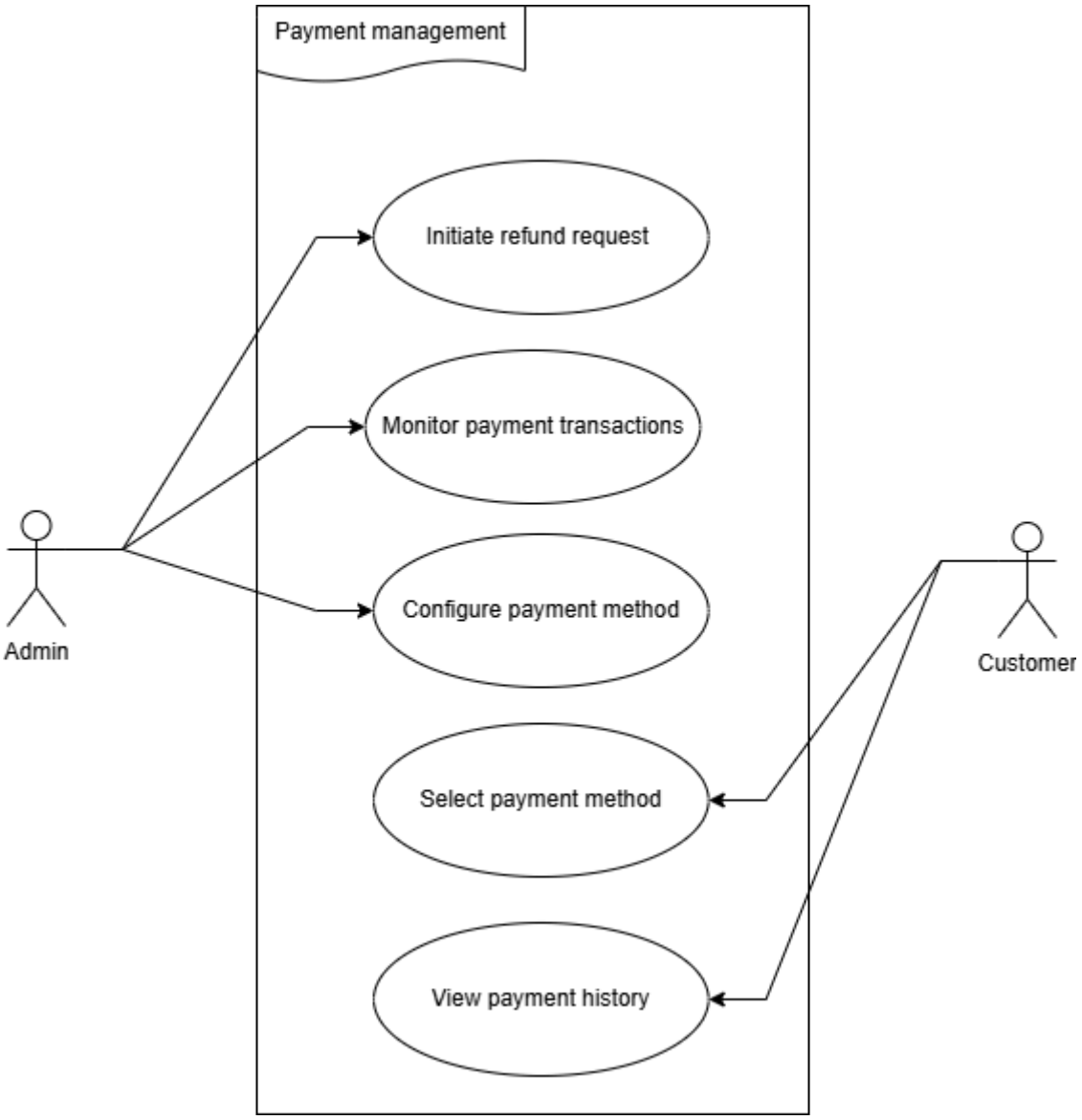


Figure 10: Payment Management use case diagram

Payment Management contains specific 5 features: Initiate refund request, Monitor Payment Transactions, Configure Payment methods, Select Payment method, View Payment history

3.3 Product & Inventory

3.3.1 Product Management

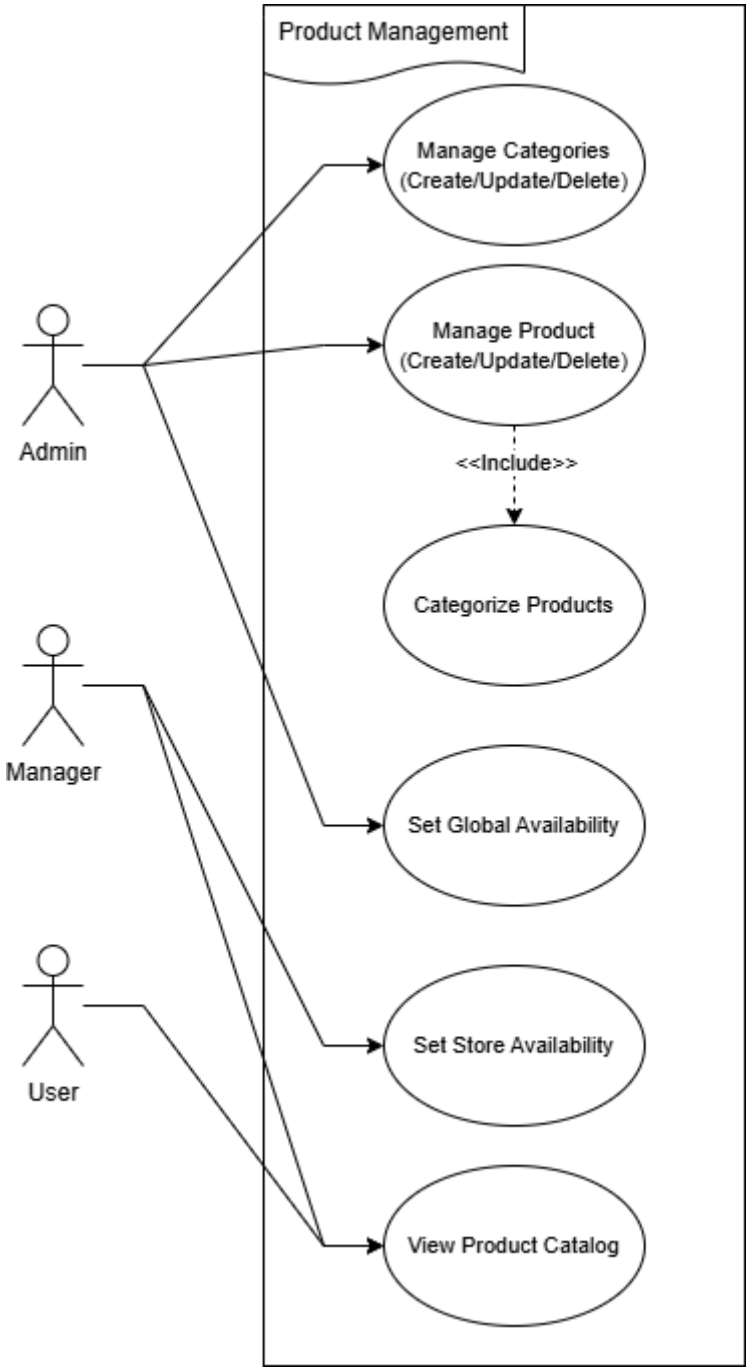


Figure 11: Product Management Service use case diagram

Product Management contains specific 12 features: Create Product, Get List of Products, View Product detail, Update Product, Categorize Product, Update Product Image, Set Product available, Delete Product, Create Category, Update Category, Delete Category, Select Ingredients for Product.

3.3.2 Inventory Management

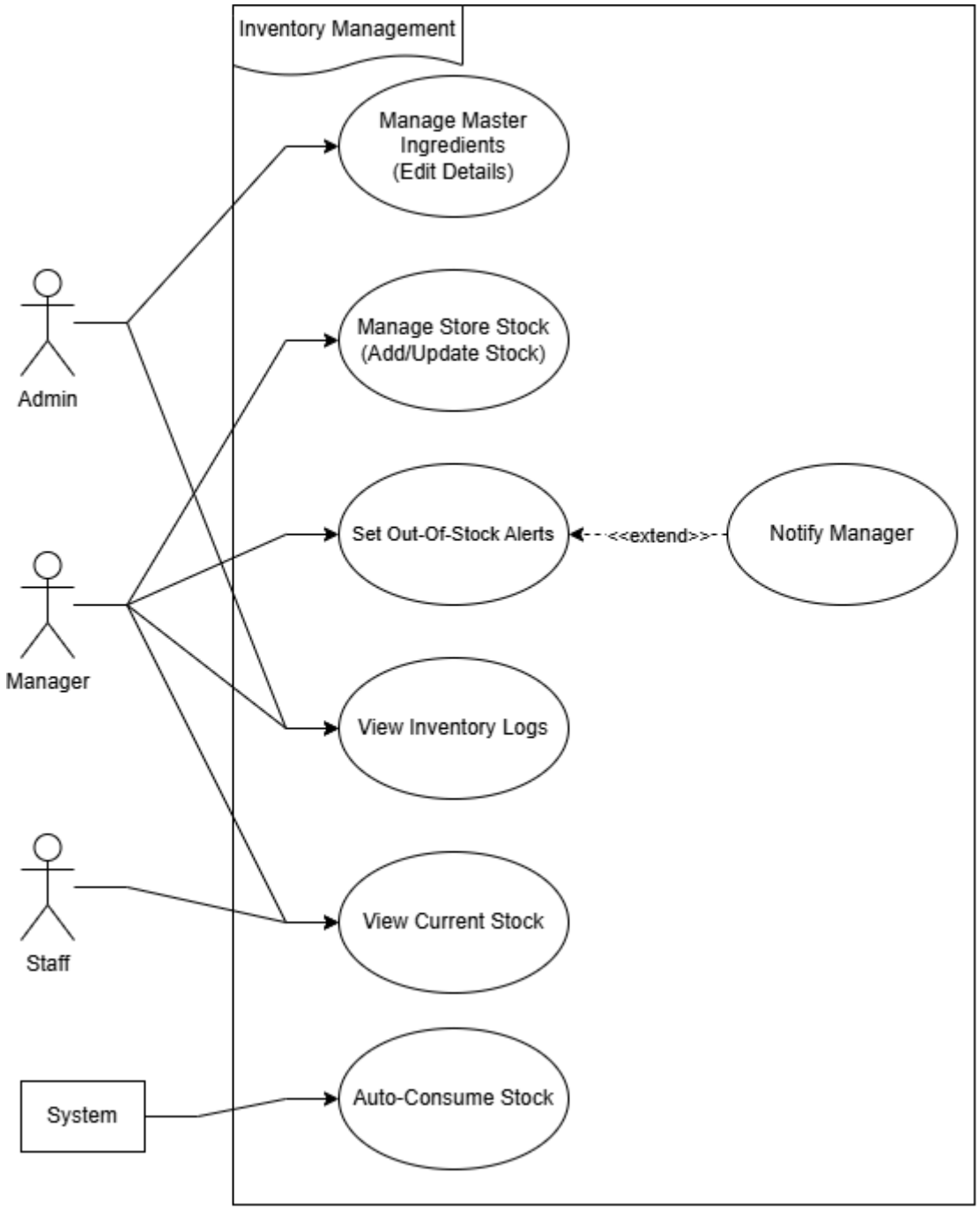


Figure 12: Inventory Management Service use case diagram

Inventory Management contains specific 6 features: Add Ingredient Stock, Edit Ingredient detail, Set out of stock alert, Deduct Ingredient for each order, View Inventory logs, View Ingredient levels.

3.4 Order & Delivery

3.4.1 Order Online

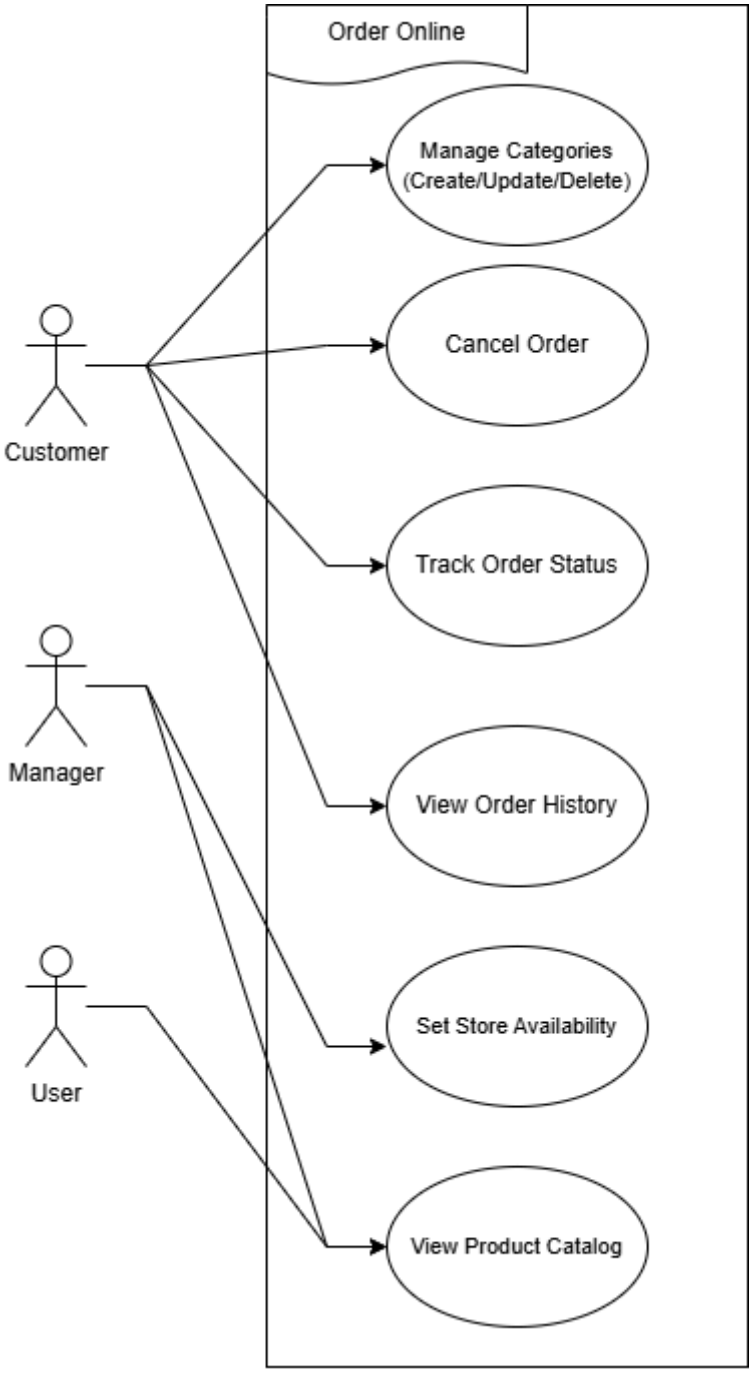


Figure 13: Order Online Management use case diagram

Order Online Management Service contains specific 7 features: Create Order, View Order detail, Cancel Order, Track Order status, Receive Order notification, Attach payment, View Order history.

3.4.2 Order Pos Management

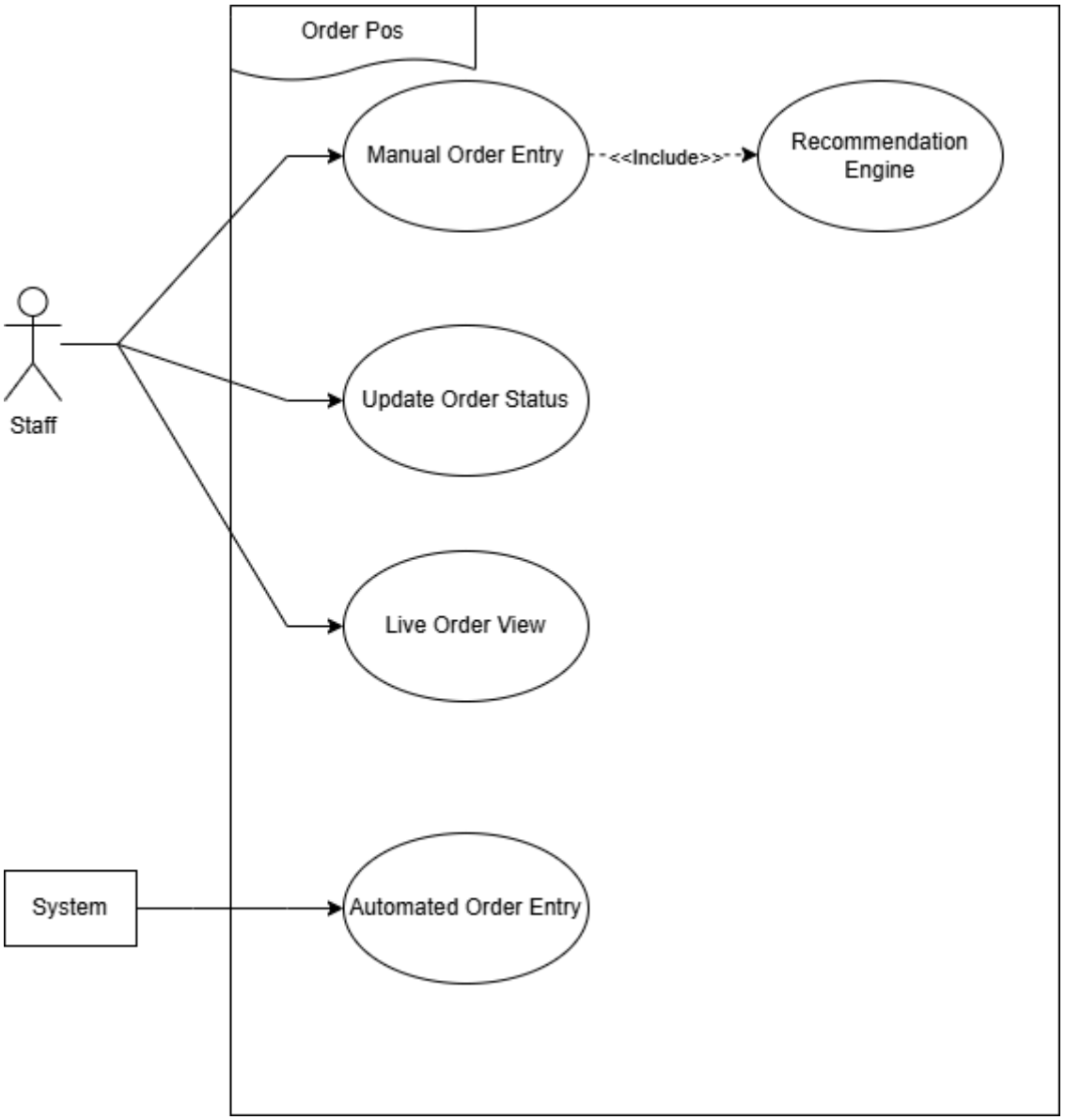


Figure 14: Order POS Management use case diagram

Order POS Management Service contains specific 10 features: POS Order entry, Update Order status, View live Orders, Estimate Preparation time, View Order list, Edit Order, Reassign Order, Flag Problematic Order, Order entry automation, Recommendation Engine.

3.4.3 Shift Management

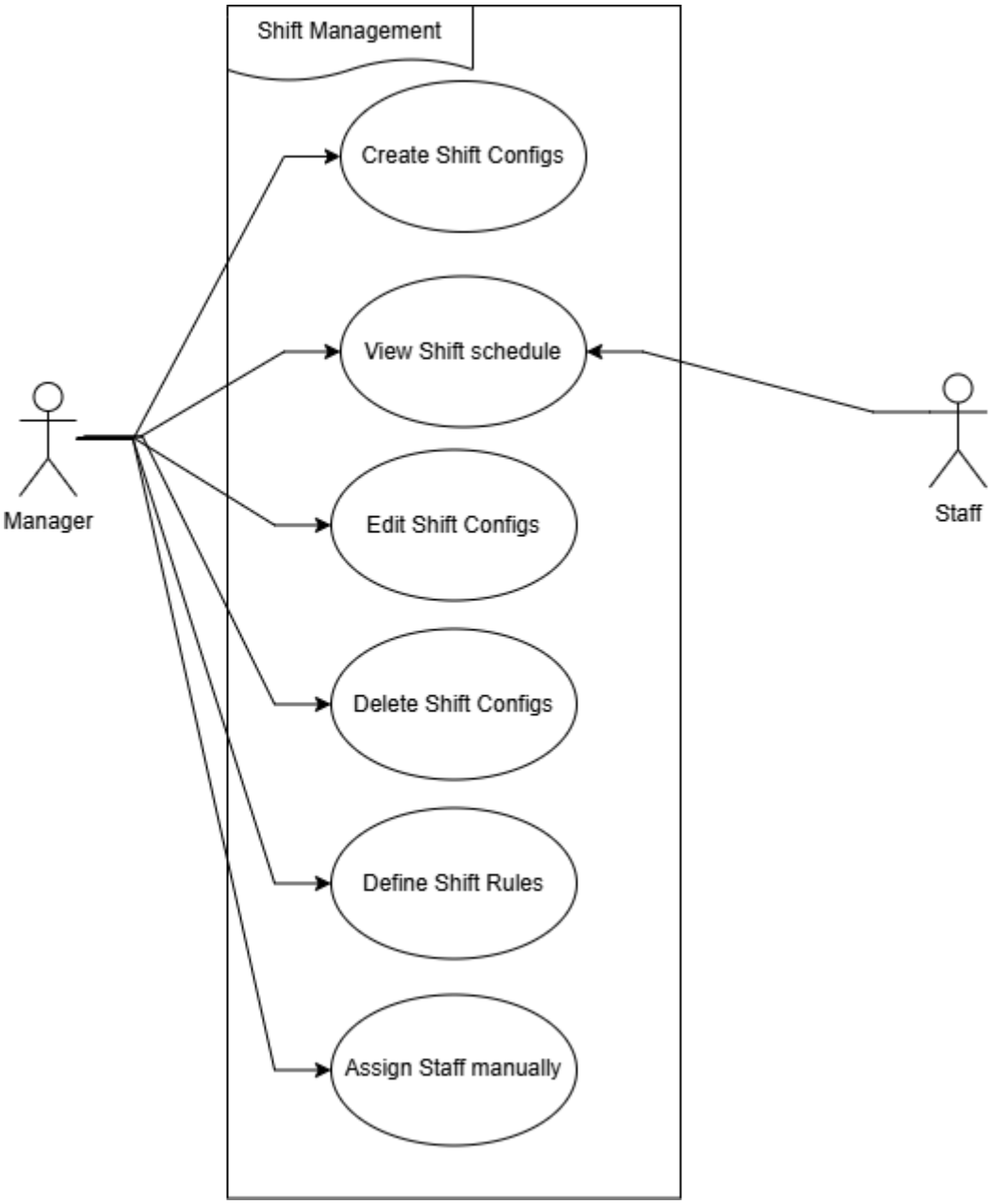


Figure 15: Shift Management use case diagram

Shift management service contains specific 6 features: Create shift configs, View shift schedule, Edit shift configs, Delete shift configs, Define shift rules, Assign staff manually

3.4.4 Delivery Management

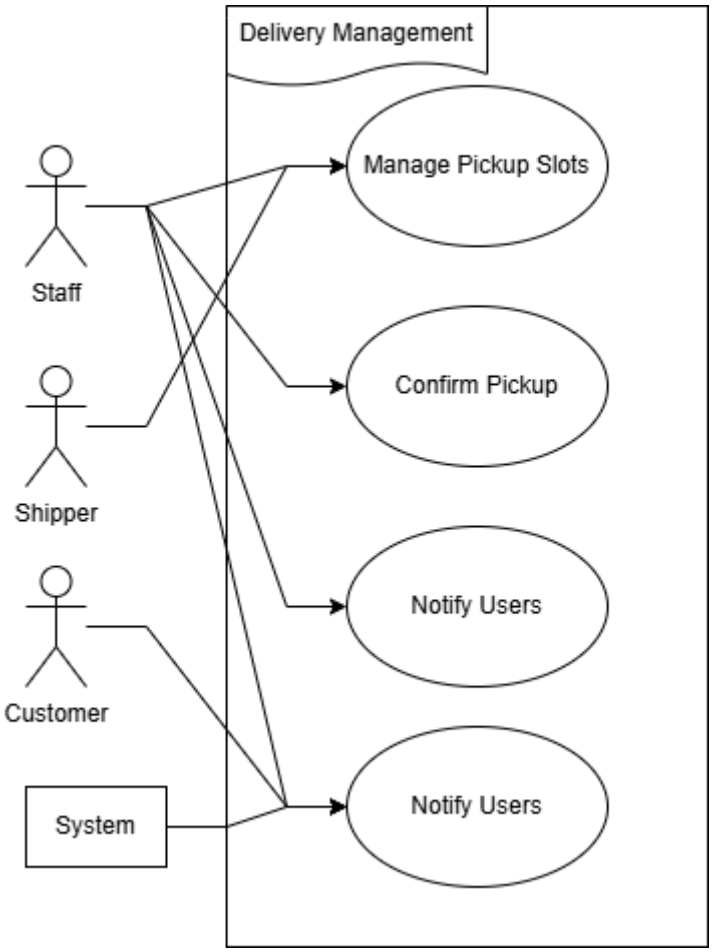


Figure 16: Delivery Management Service use case diagram

Delivery Management Service contains specific 5 features: Pickup time slots, Notify user, Confirm pickup, Assign deliver staff, Track delivery information.

4. System Overview

4.1 System Constraints

- The system must be deployable on a cost-effective hosting environment, such as shared hosting, a virtual private server, or an internal company-managed server.
- The design and implementation must adhere to widely accepted web development best practices, utilizing standard technologies such as HTML, CSS, JavaScript, and a backend framework supported by the team (e.g., Node.js, PHP, or Python).
- Each microservice (IAM, Product Management, etc.) must manage its own dedicated database instance to promote modularity and scalability.

4.2 Assumptions and Dependencies

- It is assumed that all users will access the system through modern web browsers that support current standards (e.g., Google Chrome, Mozilla Firefox, Microsoft Edge).
- The organization will provide access to internal employee information such as employee ID and email for authentication and account verification purposes.
- The current system scope excludes online payments. It is assumed that payment transactions will be handled physically at the point of pickup.

4.3 Future Enhancements

- The system may be enhanced in future iterations to support online payment integration, mobile application versions (iOS and Android), and loyalty or rewards programs for employee incentives.
- A reporting and analytics module may be developed to analyze sales trends, customer preferences, and operational performance, including peak hours and staff workload.

4.4 Legal and Regulatory Considerations

- All user interactions, order records, and associated activity logs must be securely stored and retained only for the period necessary for operational or audit purposes, in compliance with data retention policies.
- The system must follow industry-standard best practices for protecting user privacy and data integrity.

5. Non-functional Requirements

5.1 Reliability

- The system must operate consistently with minimal errors, ensuring accurate order placement, user authentication, and data storage.
- It must include automated monitoring, error detection, and failover mechanisms to handle service disruptions gracefully.
- The overall platform should achieve 99.9% uptime availability. In the event of a system failure, log continuity and error capture must remain intact via redundant logging mechanisms. Critical services such as IAM must employ redundant architecture and failover strategies to ensure uninterrupted operation.

5.2 Scalability

- The architecture must support both horizontal and vertical scaling to accommodate growing demand, increased user base, and additional feature integration.
- The system must efficiently handle increased load from concurrent orders, staff interactions, and inventory updates without performance degradation.

- IAM services must scale to accommodate growing numbers of users and authentication events, especially during peak operational hours.

5.3 Supportability

- The system must be designed for maintainability, allowing technical teams to perform updates, patches, or service extensions with minimal disruption to ongoing operations.
- Clear, up-to-date documentation must be provided to support troubleshooting, integration, and deployment workflows.

5.4 Availability

- The platform must provide high availability, ensuring that core features such as order placement and delivery tracking remain operational 24/7.
- Scheduled maintenance should be minimized and communicated in advance, and recovery procedures must be well-documented and tested.

5.5 Performance

- The system should support rapid transaction processing and UI responsiveness, allowing orders to be placed, updated, and tracked in real time.
- All authentication and authorization flows should complete within milliseconds, supporting thousands of concurrent users.
- Load balancing and caching techniques must be implemented to optimize responsiveness during high-traffic periods.
- Log ingestion and data storage processes should support high-throughput events, ensuring smooth logging without delays.

5.6 Security & Privacy

- All sensitive data, including user credentials and order logs, must be encrypted at rest and in transit using strong encryption standards (e.g., AES-256).
- Authentication protocols must support multi-factor authentication (MFA) and secure session handling.
- Continuous monitoring, alerting, and auditing must be implemented to detect, respond to, and log security events.
- The system must comply with applicable data protection regulations such as GDPR, CCPA, or ISO 27001, including provisions for data minimization, consent management, and anonymization.

5.7 Compatibility

- Any new features or services added to the system must be backward-compatible with the existing architecture and interfaces.
- Integration of external tools or APIs must conform to existing protocols and standards used across all services.

5.8 Maintainability

- The platform must support non-intrusive service upgrades and patches without requiring system-wide downtime.
- Version control, changelogs, and configuration management must be consistently applied across all modules.
- All services must be accompanied by detailed API documentation, deployment guides, and configuration instructions.